

Regulating eukaryotic gene expression with aptamers.

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Aptamers are RNA or DNA oligonucleotides identified within a randomly synthesized library, through an in vitro selection procedure. The selected candidates display a pre-determined property of interest with respect to a given target. Successful selection has been carried out against targets ranging from small (amino acids, antibiotics) to macro-molecules (proteins, nucleic acids). They generally show an affinity in the nanomolar range and a high specificity of target recognition. Interestingly, aptamers selected against purified targets in the test tube retain their properties within cells. RNA aptamers can be generated in situ from an appropriate DNA construct or delivered as nuclease-resistant oligonucleotide analogues. For example, aptamers recognizing RNA structure through loop-loop interactions modulate the trans-activation of in vitro transcription mediated by the TAR RNA element of human immunodeficiency virus type 1. Consequently, they constitute both exquisite tools for functional genomics analysis and promising prototypes of therapeutic agents. Natural aptameric motifs have been identified within mRNA sequences, which upon binding to a metabolite control the expression of the encoded gene, which is generally involved in the biosynthesis of this particular metabolite.

A universal aptameric biosensor: Multiplexed detection of small analytes via aggregated perylene-based broad-spectrum quencher.

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A universal aptameric system based on the taking advantage of double-stranded DNA/peryene diimide (dsDNA/PDI) as the signal probe was developed for multiplexed detection of small molecules. Aptamers are single-stranded DNA or RNA oligonucleotides which are selected in vitro by a process known as systematic evolution of ligands by exponential enrichment. In this work, we synthesized a new kind of PDI and reported this aggregated PDI could quench the double-stranded DNA (dsDNA)-labeled fluorophores with a high quenching efficiency. The quenching efficiencies on the fluorescence of FAM, TAMRA and Cy5 could reach to $98.3\% \pm 0.9\%$, $97.2\% \pm 0.6\%$ and $98.1\% \pm 1.1\%$, respectively. This broad-spectrum quencher was then adopted to construct a multicolor biosensor via a label-free approach. A structure-switching-triggered enzymatic recycling amplification was employed for signal amplification. High quenching efficiency combined with autocatalytic target recycling amplification afforded the biosensor with high sensitivity towards small analytes. For other targets, changing the corresponding aptamer can achieve the goal. The quencher did not interfere with the catalytic activity of nuclease. The biosensor could be manipulated with similar sensitivity no matter in pre-addition or post-addition manner. Moreover, simultaneous and multiplexed analysis of several small molecules in homogeneous solution was achieved, demonstrating its potential application in the rapid screening of multiple biotargets.

Generation of ssDNA aptamers as diagnostic tool for Newcastle avian virus.

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Aptamers are short single-stranded DNA (ssDNA), RNA or synthetic XNA molecules, which are used as a class of affinity binders recognizing target molecules with a very high affinity and specificity. The aim of this study was to generate and characterize ssDNA aptamers for the detection of Newcastle disease virus

(NDV). These aptamers were selected using systematic evolution of ligands by exponential enrichment (SELEX) in combination with quantitative high-throughput DNA sequencing. After three rounds of selections, a highly enriched ssDNA pool was sequenced, and the results were analyzed using FASTAptamer Toolkit. Sequencing reads were sorted by copy numbers and clustered into groups, according to their sequence homology. Top aptameric sequences were used to develop a sandwich enzymatic linked aptamer assay (ELAA) for rapid and sensitive detection of NDV in farm samples. The selected aptamers have an affinity within the nanomolar range, and a high specificity with no cross-reactivity towards other avian viruses. Following optimization of the sandwich ELAA method, the results demonstrated that both selected aptamers Apt_NDV01 and Apt_NDV03 with dissociation constant values of 31 nM and 78.1 nM, respectively, showed the highest specificity and affinity for NDV detection. The ELAA results were verified by quantitative real-time PCR, demonstrating strong concordance, and showing outstanding accuracy for detection of NDV in field sample. In summary, combination of SELEX with high-throughput DNA sequencing allowed rapid screening and selection of aptamers. The selected aptamers allowed recognition of NDV with high affinities. This is the first report that uses a validated sandwich ELAA for rapid and specific detection of NDV in poultry samples.

AptamerRunner: An accessible aptamer structure prediction and clustering algorithm for visualization of selected aptamers.

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Aptamers are short single-stranded DNA or RNA molecules with high affinity and specificity for targets and are generated using the iterative systematic evolution of ligands by exponential enrichment (SELEX) process. Next-generation sequencing (NGS) revolutionized aptamer selections by allowing a more comprehensive analysis of SELEX-enriched aptamers as compared to Sanger sequencing. The current challenge with aptamer NGS datasets is identifying a diverse cohort of candidate aptamers with the highest likelihood of successful experimental validation. Here we present AptamerRunner, an aptamer sequence and/or structure clustering algorithm that synergistically integrates computational analysis with visualization and expertise-directed decision making. The visual integration of networked aptamers with ranking data, such as fold enrichment or scoring algorithm results, represents a significant advancement over existing clustering tools by providing a natural context to depict groups of aptamers from which ranked or scored candidates can be chosen for experimental validation. The inherent flexibility, user-friendly design, and prospects for future enhancements with AptamerRunner have broad-reaching implications for aptamer researchers across a wide range of disciplines.

Computational approach to analyze isolated ssDNA aptamers against angiotensin II.

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Aptamers are oligonucleotides with highly structured molecules that can bind to their targets through specific 3-D conformation. Commonly, not all the nucleotides such as primer binding fixed region and some other sequences are vital for aptamers folding and interaction. Elimination of unnecessary regions needs trustworthy prediction tools to reduce experimental efforts and errors. Here we introduced a manipulated in-silico approach to predict the 3-D structure of aptamers and their target interactions. To design an approach for computational analysis of isolated ssDNA aptamers (FLC112, FLC125 and their

truncated core region including CRC112 and CRC125), their secondary and tertiary structures were modeled by Mfold and RNA composer respectively. Output PDB files were modified from RNA to DNA in the discovery studio visualizer software. Using ZDOCK server, the aptamer-target interactions were predicted. Finally, the interaction scores were compared with the experimental results. In-silico interaction scores and the experimental outcomes were in the same descending arrangement of FLC112>CRC125>CRC112>FLC125 with similar intensity. The consistent results of innovative in-silico method with experimental outputs, affirmed that the present method may be a reliable approach. Also, it showed that the exact in-silico predictions can be utilized as a credible reference to find aptameric fragments binding potency.

Development of Phosphorothioate DNA and DNA Thioaptamers.

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Nucleic acid aptamers are short RNA- or DNA-based affinity reagents typically selected from combinatorial libraries to bind to a specific target such as a protein, a small molecule, whole cells or even animals. Aptamers have utility in the development of diagnostic, imaging and therapeutic applications due to their size, physico-chemical nature and ease of synthesis and modification to suit the application. A variety of oligonucleotide modifications have been used to enhance the stability of aptamers from nuclease degradation in vivo. The non-bridging oxygen atoms of the phosphodiester backbones of RNA and DNA aptamers can be substituted with one or two sulfur atoms, resulting in thioaptamers with phosphorothioate or phosphorodithioate linkages, respectively. Such thioaptamers are known to have increased binding affinity towards their target, as well as enhanced resistance to nuclease degradation. In this review, we discuss the development of phosphorothioate chemistry and thioaptamers, with a brief review of selection methods.

Selection and Application of Aptamers and Intramers.

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Aptamers are auspicious nucleic acid ligands for targeting different molecules, such as small molecules, peptides, proteins, or even whole living cells. They are short single-stranded DNA or RNA oligonucleotides, which can fold into complex three-dimensional structures and bind selectively their targets. Using the combinatorial chemistry process SELEX (Systematic Evolution of Ligands by EXponential Enrichment), target specific aptamers can be selected. These aptamers have a variety of application possibilities and can be used as sensors, diagnostic, imaging or therapeutic agents, and in the field of regenerative medicine for tissue engineering.

Nucleic acid aptamers as adjuncts to vaccine development.

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Nucleic acid 'aptamers', a term derived from the Latin word aptus, 'to fit', are RNA or DNA oligonucleotides that conform to the three-dimensional structure of a selected protein, peptide or small

molecules' functional moiety. The 'lock and key' relationship between aptamers and their binding partner permits distinction between closely related but non-identical members of a protein family, or between different functional or conformational states of the same protein. This, along with other properties, separates aptamers from antibodies--the most popular class of molecular recognition tool for the past three decades. Despite the chemical, biological and manufacturing advantages offered by nucleic acid aptamers in a wide variety of conditions, and their generation against a range of clinically relevant targets, including growth factors, transcription factors and coagulation proteins, by two dozen or more companies devoted to the technology platform, only one aptamer, developed for the treatment of wet age-related macular degeneration, is currently available for use in humans. Nevertheless, phase I and II clinical trials for several indications are proceeding with considerable enthusiasm. The potential application of nucleic acid aptamers in novel arenas, including molecular imaging, vaccine development, immunomodulation, decoys for natural RNA-binding events, antiviral therapeutics and both cancer prophylaxis and treatment, is emerging with a pioneering mentality destined to change the paradigm of patient care.

Clinical applications of nucleic acid aptamers in cancer.

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Nucleic acid aptamers are small single-stranded DNA or RNA oligonucleotide segments, which bind to their targets with high affinity and specificity via unique three-dimensional structures. Aptamers are generated by an iterative in vitro selection process, termed as systematic evolution of ligands by exponential enrichment. Owing to their specificity, non-immunogenicity, non-toxicity, easily modified chemical structure and wide range of targets, aptamers appear to be ideal candidates for various clinical applications (diagnosis or treatment), such as cell detection, target diagnosis, molecular imaging and drug delivery. Several aptamers have entered the clinical pipeline for applications in diseases such as macular degeneration, coronary artery bypass graft surgery and various types of cancer. The aim of this review was to summarize and highlight the clinical applications of aptamers in cancer diagnosis and treatment.

Cell-specific aptamer-mediated targeted drug delivery.

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Nucleic acid aptamers are in vitro-selected small, single-stranded DNA or RNA oligonucleotides that can specifically recognize their target on the basis of their unique 3-dimensional structures. Recent advances in the development of escort aptamers to deliver and enhance the efficacy of other therapeutic agents have drawn enthusiasm in exploiting cell-type-specific aptamers as drug delivery vehicles. This review mainly focuses on the recent developments of aptamer-mediated targeted delivery systems. We also place particular emphasis on aptamers evolved against cell membrane receptors and possibilities for translation to clinical applications.